

CHAPTER 6: THERMODYNAMICS (JEE FOCUS)

Topics Covered:

- System & surroundings
- Types of systems
- Internal energy
- First law of thermodynamics
- Enthalpy
- Thermochemical equations
- Hess's law
- Entropy & spontaneity
- Gibbs free energy

Core Concepts:

- System → part under study
- Open, closed, isolated systems
- State function → depends only on state
- Path function → depends on path
- Internal energy (U) → total energy
- First law → energy conserved
- $\Delta U = q + w$
- Work → $w = -P\Delta V$
- Enthalpy (H) = $U + PV$
- $\Delta H = q$ (constant pressure)
- Exothermic → $\Delta H < 0$
- Endothermic → $\Delta H > 0$
- Hess law → enthalpy additive
- Entropy (S) → randomness
- $\Delta S > 0$ → more disorder
- Spontaneity → depends on ΔG
- $\Delta G = \Delta H - T\Delta S$

Key Formulas:

- $\Delta U = q + w$
- $w = -P\Delta V$
- $H = U + PV$
- $\Delta H = \Delta U + \Delta nRT$
- $\Delta G = \Delta H - T\Delta S$
- $q = mc\Delta T$

Important Patterns:

- State functions → path independent
- Exothermic → heat released
- $\Delta G < 0$ → spontaneous
- $\Delta G > 0$ → non-spontaneous
- $\Delta G = 0$ → equilibrium
- Entropy $\uparrow$ → spontaneity $\uparrow$
- Hess law → break into steps
- Sign convention → very important

Common Mistakes:

- Wrong sign of work
- Confusing ΔH and ΔU
- Ignoring ΔnRT term
- Mistakes in ΔG sign
- Mixing state vs path functions
- Wrong unit conversions
- Forgetting temperature in Kelvin

Examples (Important Cases):

- Calculate work in expansion
- ΔH using Hess law
- Spontaneity using ΔG
- Exo vs endothermic reactions
- Entropy change cases

Exam Tips:

- Always check sign convention
- Convert temperature to Kelvin
- Use Hess law smartly
- Focus on ΔG questions
- Practice numericals
- Write formulas before solving
- Avoid sign mistakes